

UNIVERSITY OF BRISTOL

Faculty of Science

MOCK EXAMINATION — Paper 6 (Easy)**Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Duration: 3 Hours

Total Marks: 100

Instructions:

- Answer **ALL** questions.
 - **Section A** (Oscillations and Waves): 10 multiple-choice questions, 20 marks total.
 - **Section B** (Quantum Physics): 10 multiple-choice questions, 20 marks total.
 - **Section C**: 4 long-answer questions, 60 marks total.
 - An approved calculator is permitted.
 - A formula sheet is provided at the end of this paper.
 - Show all working for Section C. Marks will be awarded for clear, logical solutions.
-

Section A: Oscillations and Waves

Answer all 10 questions. Each question is worth 2 marks.

1. A particle undergoes simple harmonic motion (SHM). Which of the following correctly describes the restoring force acting on the particle? [2 marks]

- A. The force is constant in magnitude and always directed toward the equilibrium position.
- B. The force is proportional to the displacement from equilibrium and directed toward the equilibrium position: $F = -kx$.
- C. The force is proportional to the square of the displacement from equilibrium.
- D. The force is proportional to the velocity of the particle.

2. A mass of $m = 0.5$ kg is attached to a spring with spring constant $k = 200$ N/m. What is the period of oscillation? [2 marks]

- A. $T = 0.157$ s
- B. $T = 0.201$ s
- C. $T = 0.314$ s
- D. $T = 0.628$ s

3. A mass oscillates in SHM with amplitude $A = 0.10$ m and angular frequency $\omega_0 = 5$ rad/s. What is the maximum speed of the mass? [2 marks]

- A. 0.25 m/s
- B. 0.50 m/s
- C. 2.5 m/s
- D. 5.0 m/s

4. In a damped oscillating system, the quality factor Q is defined as: [2 marks]

- A. $Q = \gamma/\omega_0$, the ratio of the damping constant to the natural frequency.
- B. $Q = \omega_0/\gamma$, the ratio of the natural angular frequency to the damping constant.
- C. $Q = \omega_0\gamma$, the product of the natural frequency and damping constant.
- D. $Q = \omega_d/\omega_0$, the ratio of the damped frequency to the natural frequency.

5. A wave has frequency $f = 500$ Hz and wavelength $\lambda = 0.68$ m. What is the speed of the wave? [2 marks]

- A. 170 m/s

- B. 340 m/s
C. 500 m/s
D. 735 m/s
6. A string of length L is fixed at both ends and vibrates in its 3rd harmonic. How many antinodes are present on the string? [2 marks]
A. 1
B. 2
C. 3
D. 4
7. The sound intensity level of 0 dB corresponds to a sound intensity of: [2 marks]
A. 0 W/m²
B. 1 W/m²
C. 10⁻¹² W/m²
D. 10⁻⁶ W/m²
8. A sound source moves *toward* a stationary observer at speed v_s (where $v_s < v$, the speed of sound). The frequency heard by the observer is: [2 marks]
A. Lower than the emitted frequency.
B. Equal to the emitted frequency.
C. Higher than the emitted frequency.
D. Zero, because the source is moving.
9. Two tuning forks produce sound waves with frequencies $f_1 = 256$ Hz and $f_2 = 260$ Hz. What is the beat frequency heard when both sound simultaneously? [2 marks]
A. 2 Hz
B. 4 Hz
C. 258 Hz
D. 516 Hz
10. The period of a simple pendulum undergoing small oscillations depends on which of the following quantities? [2 marks]
A. The mass of the bob and the length of the string only.
B. The length of the string and the acceleration due to gravity only.
C. The amplitude of oscillation and the mass of the bob only.
D. The mass of the bob, the length of the string, and the acceleration due to gravity.

Section B: Quantum Physics

Answer all 10 questions. Each question is worth 2 marks.

1. In Dirac notation, the symbol $|\psi\rangle$ represents: [2 marks]
 - A. A probability
 - B. A complex number
 - C. A quantum state (state vector/ket)
 - D. An operator
2. In Dirac notation, the quantity $\langle\phi|\psi\rangle$ is: [2 marks]
 - A. A ket (column vector)
 - B. A bra (row vector)
 - C. An operator (matrix)
 - D. A complex scalar (complex number)
3. For two normalised quantum states $|\phi\rangle$ and $|\psi\rangle$, the quantity $|\langle\phi|\psi\rangle|^2$ represents: [2 marks]
 - A. The energy of the state $|\psi\rangle$
 - B. The probability of finding the system in state $|\phi\rangle$ given that it is in state $|\psi\rangle$
 - C. The expectation value of the Hamiltonian
 - D. The phase difference between the two states
4. In the eigenvalue equation $\hat{A}|a\rangle = a|a\rangle$, the quantity a is called: [2 marks]
 - A. The expectation value
 - B. The normalisation constant
 - C. The eigenvalue
 - D. The eigenvector
5. For a spin- $\frac{1}{2}$ particle, the possible results of measuring S_z are: [2 marks]
 - A. 0 and $\hbar$
 - B. $+\hbar/2$ and $-\hbar/2$
 - C. $+\hbar$ and $-\hbar$
 - D. +1 and -1

6. A spin- $\frac{1}{2}$ particle is in the state $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$, where α and β are complex numbers. For this state to be normalised, which condition must hold? [2 marks]

- A. $\alpha + \beta = 1$
- B. $\alpha^2 + \beta^2 = 1$
- C. $|\alpha|^2 + |\beta|^2 = 1$
- D. $|\alpha| + |\beta| = 1$

7. An operator $\hat{A}$ is said to be Hermitian if: [2 marks]

- A. $\hat{A}^2 = \hat{I}$
- B. $\hat{A}^\dagger = -\hat{A}$
- C. $\hat{A}^\dagger = \hat{A}$
- D. $\hat{A}^\dagger \hat{A} = \hat{I}$

8. A particle with spin quantum number $s = 1$ has how many distinct spin states (i.e. distinct values of m_s)? [2 marks]

- A. 1
- B. 2
- C. 3
- D. 4

9. The proton has quark content uud . Given that the up quark has charge $+\frac{2}{3}e$ and the down quark has charge $-\frac{1}{3}e$, what is the total charge of the proton? [2 marks]

- A. 0
- B. $+\frac{1}{3}e$
- C. $+e$
- D. $+2e$

10. An electron is classified as which type of particle? [2 marks]

- A. A boson and a baryon
- B. A fermion and a lepton
- C. A boson and a lepton
- D. A fermion and a baryon

Section C: Long Answer Questions

Answer all 4 questions. Show all working.

Question 1.

[15 marks]

SUVAT, Forces, and Energy — Direct Application

- (a) A ball is thrown vertically upward from ground level with an initial speed of 20 m/s. Take $g = 9.81 \text{ m/s}^2$.
- (i) Find the time taken to reach the maximum height. 1
 - (ii) Find the maximum height reached. 1
 - (iii) Find the total time of flight (i.e. the time for the ball to return to ground level). 1
- (b) A 10 kg box is pushed across a horizontal floor by a horizontal force of 60 N. The coefficient of kinetic friction between the box and the floor is $\mu_k = 0.4$.
- (i) List all the forces acting on the box and describe the direction of each (i.e. describe the free body diagram). 1
 - (ii) Calculate the kinetic friction force. 1
 - (iii) Find the acceleration of the box. 1
- (c) A 2 kg ball is dropped from a height of 5 m above the ground. It reaches the ground and then slides onto a frictionless horizontal surface where it encounters a spring with spring constant $k = 500 \text{ N/m}$. Assume no energy is lost.
- (i) What is the speed of the ball just before it reaches the ground? 2
 - (ii) Using energy conservation, find the maximum compression of the spring. 2
- (d) A 3 kg object moving at 4 m/s to the right collides with a 1 kg object that is initially at rest.
- (i) If the collision is perfectly inelastic (the objects stick together), find the final velocity of the combined mass. 2
 - (ii) Calculate the kinetic energy lost in the perfectly inelastic collision. 1
 - (iii) If instead the collision is elastic, use the elastic collision formula from the formula sheet to find the final velocity of the 1 kg object. 2

Question 2.

[15 marks]

Heat, Ideal Gas, and Basic Thermodynamics

- (a) 0.5 kg of water at 20 °C is heated to 100 °C and then completely converted to steam at 100 °C. Calculate the total energy required. Use $c_{\text{water}} = 4200 \text{ J/(kg K)}$ and $L_{\text{vap}} = 2.26 \times 10^6 \text{ J/kg}$. 3
- (b) A container holds 1 mol of ideal monatomic gas at 300 K and $1.0 \times 10^5 \text{ Pa}$.
- (i) Find the volume of the gas. 1
 - (ii) The gas is heated at constant pressure until its temperature reaches 600 K. Find the new volume. 1
 - (iii) Calculate the work done by the gas during this isobaric expansion. 1
 - (iv) Calculate the heat absorbed by the gas. 1
- (c) The gas from part (b), now at 600 K and pressure $1.0 \times 10^5 \text{ Pa}$, undergoes an adiabatic expansion to twice its volume. Take $\gamma = 5/3$.
- (i) Find the final temperature after the adiabatic expansion. 2
 - (ii) Find the final pressure. Is it higher or lower than the initial pressure $1.0 \times 10^5 \text{ Pa}$? 2
- (d) A Carnot engine operates between a hot reservoir at $T_h = 500 \text{ K}$ and a cold reservoir at $T_c = 300 \text{ K}$. The engine absorbs $Q_h = 1000 \text{ J}$ of heat per cycle from the hot reservoir.
- (i) What is the efficiency of the engine? 1
 - (ii) How much work does the engine do per cycle? 1
 - (iii) How much heat is rejected to the cold reservoir per cycle? 1
 - (iv) What is the entropy change of the hot reservoir per cycle? Of the cold reservoir? Of the universe? 1

Question 3.

[15 marks]

Direct SHM and Wave Applications

- (a) A mass $m = 0.25$ kg is attached to a spring with spring constant $k = 100$ N/m and oscillates with amplitude $A = 4$ cm = 0.04 m.
- (i) Find the angular frequency ω_0 and the period T . 2
 - (ii) Find the total energy of the oscillator. 1
 - (iii) Find the maximum speed of the mass. 1
- (b) A transverse wave on a string is described by $y(x, t) = 0.04 \sin(3x - 12t)$ m, where x is in metres and t is in seconds.
- (i) Determine the amplitude, wavelength, frequency, and wave speed. 2
 - (ii) In which direction is the wave travelling? 1
- (c) A string of length $L = 2$ m is fixed at both ends. The wave speed on the string is $v = 200$ m/s.
- (i) Find the fundamental frequency f_1 . 1
 - (ii) Find the frequencies of the 2nd and 3rd harmonics. 1
 - (iii) Sketch the displacement patterns for the first three harmonics on the string, clearly indicating nodes (N) and antinodes (A). 2
- (d) A fire engine siren emits sound at $f = 800$ Hz. The speed of sound is $v = 340$ m/s.
- (i) The fire engine approaches a stationary observer at $v_s = 30$ m/s. What frequency does the observer hear? 2
 - (ii) The fire engine passes and moves away from the observer at $v_s = 30$ m/s. What frequency does the observer now hear? 2

Question 4.

[15 marks]

Basic Dirac Notation and Spin Measurements

- (a) A spin- $\frac{1}{2}$ particle is in the state

$$|\psi\rangle = \frac{3}{5} |\uparrow_z\rangle + \frac{4}{5} |\downarrow_z\rangle.$$

- (i) Verify that the state is normalised. 1
 - (ii) If S_z is measured, what are the possible outcomes and their probabilities? 2
- (b) Using the spin eigenstates from the formula sheet, express $|\uparrow_z\rangle$ and $|\downarrow_z\rangle$ in terms of $|\uparrow_x\rangle$ and $|\downarrow_x\rangle$. Hence write $|\psi\rangle$ from part (a) in the x -basis and find the probability of measuring $S_x = +\hbar/2$. 4
- (c) Consider the operator $\hat{A} = |\uparrow\rangle\langle\uparrow| - |\downarrow\rangle\langle\downarrow|$ (written in the z -basis).
- (i) Write $\hat{A}$ as a 2×2 matrix in the z -basis. 1
 - (ii) What are the eigenvalues and eigenvectors of $\hat{A}$? 2
 - (iii) Is $\hat{A}$ Hermitian? How do you know? 1
- (d) The state $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$ is an eigenstate of which spin operator (S_x , S_y , or S_z)? What is the corresponding eigenvalue? If S_z is measured on this state, what is the expectation value $\langle S_z \rangle$? 4

Formula Sheet

Mathematical Formulae

Binomial expansion: $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots \quad (|x| \ll 1)$

Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$

Trig identities: $\sin A + \sin B = 2 \cos\left(\frac{A-B}{2}\right) \sin\left(\frac{A+B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$

Mechanics

SUVAT: $v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u+v)t$

Dot product: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta = a_x b_x + a_y b_y + a_z b_z$

Cross product: $\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$

Rocket equation: $\Delta v = u \ln\left(\frac{m_0}{m_f}\right)$

Gravitational PE: $U = -\frac{GMm}{r}$

Escape velocity: $v_e = \sqrt{\frac{2GM}{R}}$

Force from PE: $F = -\frac{dU}{dx}$

Elastic collision (1D): $v'_1 = \frac{m_1 - m_2}{m_1 + m_2}v_1 + \frac{2m_2}{m_1 + m_2}v_2$

Centre of mass: $\mathbf{R}_{\text{cm}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$

Reduced mass: $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Centripetal accel: $a_c = \frac{v^2}{r} = \omega^2 r$

Parallel axis thm: $I = I_{\text{cm}} + Md^2$

Atwood machine: $a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$

Moments of inertia (about axis through centre of mass):

Thin ring (radius R):	$I = MR^2$
Solid disk (radius R):	$I = \frac{1}{2}MR^2$
Solid sphere (radius R):	$I = \frac{2}{5}MR^2$
Hollow sphere (radius R):	$I = \frac{2}{3}MR^2$
Thin rod (length L , about centre):	$I = \frac{1}{12}ML^2$
Thin rod (length L , about end):	$I = \frac{1}{3}ML^2$

Properties of Matter

Ideal gas: $PV = nRT$

Heat capacities: $C_P - C_V = R$ (per mole)

Adiabatic: $PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const}$

Isothermal work: $W = nRT \ln\left(\frac{V_2}{V_1}\right)$

Carnot efficiency: $\eta = 1 - \frac{T_c}{T_h}$

Entropy: $dS = \frac{dQ_{\text{rev}}}{T}, \quad \Delta S = mc \ln\left(\frac{T_2}{T_1}\right), \quad \Delta S = nR \ln\left(\frac{V_2}{V_1}\right)$

Lennard-Jones: $V(r) = 4\varepsilon \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^6 \right]$

Mean KE: $\langle KE \rangle = \frac{3}{2}k_B T$

MB speeds: $c_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}, \quad c_{\text{avg}} = \sqrt{\frac{8k_B T}{\pi m}}, \quad c_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Oscillations and Waves

SHM:	$x(t) = A \cos(\omega_0 t + \phi), \quad \omega_0 = \sqrt{k/m}$
Damped SHM:	$x(t) = A e^{-\gamma t/2} \cos(\omega_d t + \phi), \quad \omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$
Quality factor:	$Q = \omega_0/\gamma$
Forced amplitude:	$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}}$
Phase lag:	$\tan \delta = \frac{\gamma \omega}{\omega_0^2 - \omega^2}$
Pendulums:	$T_{\text{simple}} = 2\pi \sqrt{L/g}, \quad T_{\text{physical}} = 2\pi \sqrt{I/(Mgh)}$
Wave equation:	$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$
String speed:	$v = \sqrt{T/\mu}$
Sound in gas:	$v = \sqrt{\gamma P/\rho}$
Harmonic wave:	$y = A \sin(kx - \omega t)$
String impedance:	$Z = \mu v = \sqrt{\mu T}$
Reflection:	$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \quad t = \frac{2Z_1}{Z_1 + Z_2}$
Power (string):	$\langle P \rangle = \frac{1}{2} \mu \omega^2 A^2 v$
Decibels:	$\beta = 10 \log_{10}(I/I_0), \quad I_0 = 10^{-12} \text{ W/m}^2$
Doppler:	$f' = f \frac{v \pm v_o}{v \mp v_s}$
Group velocity:	$v_g = \frac{d\omega}{dk}$
String harmonics:	$f_n = \frac{nv}{2L} \quad (n = 1, 2, 3, \dots)$
Open-closed pipe:	$f_n = \frac{nv}{4L} \quad (n = 1, 3, 5, \dots)$

Quantum Physics

Schrödinger eq: $i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$

Time evolution: $|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle, \quad U(t) = e^{-iHt/\hbar}$

Spin operators:

$$S_z = \frac{\hbar}{2} (\uparrow\uparrow\langle\uparrow - \downarrow\downarrow\langle\downarrow)$$

$$S_x = \frac{\hbar}{2} (\uparrow\uparrow\langle\downarrow + \downarrow\downarrow\langle\uparrow)$$

$$S_y = \frac{\hbar}{2} (-i \uparrow\uparrow\langle\downarrow + i \downarrow\downarrow\langle\uparrow)$$

Spin eigenstates:

$$\uparrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + \downarrow_z\rangle), \quad \downarrow_x\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - \downarrow_z\rangle)$$

$$\uparrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle + i \downarrow_z\rangle), \quad \downarrow_y\rangle = \frac{1}{\sqrt{2}} (\uparrow_z\rangle - i \downarrow_z\rangle)$$

Pauli matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Physical Constants

Speed of light	c	3.00×10^8 m/s
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck's constant	$\hbar$	1.055×10^{-34} J·s
Boltzmann constant	k_B	1.381×10^{-23} J/K
Gas constant	R	8.314 J/(mol·K)
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
Gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Acceleration due to gravity	g	9.81 m/s ²
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Elementary charge	e	1.602×10^{-19} C
Mass of Earth	$M_\oplus$	5.97×10^{24} kg
Radius of Earth	$R_\oplus$	6.37×10^6 m